

Solutions: Calculus

ECON 441: Introduction to Mathematical Economics

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Exercise 6.2

2. $y = 5x^2 - 4x$

(a)

$$\begin{aligned}\frac{\Delta y}{\Delta x} &= \frac{f(x + \Delta x) - f(x)}{\Delta x} \\ &= \frac{5(x + \Delta x)^2 - 4(x + \Delta x) - (5x^2 - 4x)}{\Delta x} \\ &= \frac{5(x^2 + 2x\Delta x + (\Delta x)^2) - 4x - 4\Delta x - 5x^2 + 4x}{\Delta x} \\ &= \frac{10x\Delta x + 5(\Delta x)^2 - 4\Delta x}{\Delta x} = 10x - 4 + 5\Delta x\end{aligned}$$

(b)

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} (10x - 4 + 5\Delta x) = 10x - 4$$

(c) $f'(2) = 10 \times 2 - 4 = 16$ and $f'(3) = 10 \times 3 - 4 = 26$.

3. $y = 5x - 2$

(a)

$$\frac{\Delta y}{\Delta x} = \frac{5(x + \Delta x) - 2 - (5x - 2)}{\Delta x} = \frac{5\Delta x}{\Delta x} = 5$$

It is a constant function.

(b) No: the difference quotient equals 5 whatever the size of Δx , so its limit as $\Delta x \rightarrow 0$ is also 5, i.e. $dy/dx = 5$.

Exercise 7.1

3. (a) $f'(x) = 18$, so $f'(1) = 18$ and $f'(2) = 18$.

(b) $f'(x) = 3cx^2$, so $f'(1) = 3c$ and $f'(2) = 12c$.

(c) $f'(x) = 10x^{-3}$, so $f'(1) = 10$ and $f'(2) = \frac{10}{8} = \frac{5}{4}$.

(d) $f'(x) = x^{1/3} = \sqrt[3]{x}$, so $f'(1) = 1$ and $f'(2) = \sqrt[3]{2}$.

(e) $f'(w) = 2w^{-2/3}$, so $f'(1) = 2$ and $f'(2) = 2 \cdot 2^{-2/3} = 2^{1/3}$.

(f) $f'(w) = \frac{1}{2}w^{-7/6}$, so $f'(1) = \frac{1}{2}$ and $f'(2) = \frac{1}{2} \cdot 2^{-7/6} = 2^{-13/6}$.

Exercise 7.2

3. (d) Let $f(x) = ax - b$ and $g(x) = cx^2$, so $f'(x) = a$ and $g'(x) = 2cx$.

$$\begin{aligned}\frac{d}{dx} [f(x)g(x)] &= f'(x)g(x) + f(x)g'(x) \\ &= acx^2 + (ax - b)2cx \\ &= acx^2 + 2acx^2 - 2bcx \\ &= 3acx^2 - 2bcx\end{aligned}$$

(e) Let $f(x) = (2 - 3x)(1 + x)$ and $g(x) = x + 2$, so $g'(x) = 1$. Using the product rule on $f(x)$ itself:

$$f'(x) = -3(1 + x) + (2 - 3x)(1) = -(6x + 1)$$

Then

$$\begin{aligned}\frac{d}{dx} [f(x)g(x)] &= f'(x)g(x) + f(x)g'(x) \\ &= -(6x + 1)(x + 2) + (2 - 3x)(1 + x) \\ &= -6x^2 - 13x - 2 + 2 - x - 3x^2 \\ &= -9x^2 - 14x = -x(9x + 14)\end{aligned}$$

7. (a) With $f(x) = x^2 + 3$, $f'(x) = 2x$, and $g(x) = x$, $g'(x) = 1$:

$$\begin{aligned}\frac{d}{dx} \frac{f(x)}{g(x)} &= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} \\ &= \frac{(2x)x - (x^2 + 3) \cdot 1}{x^2} = \frac{x^2 - 3}{x^2}\end{aligned}$$

(b)

$$\frac{d}{dx} \frac{x + 9}{x} = \frac{1 \cdot x - 1 \cdot (x + 9)}{x^2} = \frac{-9}{x^2}$$

(c)

$$\frac{d}{dx} \frac{6x}{x + 5} = \frac{6(x + 5) - 1 \cdot 6x}{(x + 5)^2} = \frac{30}{(x + 5)^2}$$

(d)

$$\begin{aligned}
 \frac{d}{dx} \frac{ax^2 + b}{cx + d} &= \frac{2ax(cx + d) - (ax^2 + b)c}{(cx + d)^2} \\
 &= \frac{2acx^2 + 2adx - acx^2 - bc}{(cx + d)^2} \\
 &= \frac{acx^2 + 2adx - bc}{(cx + d)^2}
 \end{aligned}$$

8. $f(x) = ax + b$

(a) $f'(x) = a$

(b) $\frac{d}{dx}(ax^2 + bx) = 2ax + b$

(c)

$$\frac{d}{dx} \frac{1}{ax + b} = \frac{0 \cdot (ax + b) - a \cdot 1}{(ax + b)^2} = \frac{-a}{(ax + b)^2}$$

(d)

$$\frac{d}{dx} \frac{ax + b}{x} = \frac{ax - (ax + b) \cdot 1}{x^2} = \frac{-b}{x^2}$$

Exercise 7.3

1.

$$\frac{dy}{du} = 3u^2 + 2 \quad \frac{du}{dx} = -2x$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (3u^2 + 2)(-2x) = -2x \left[3(5 - x^2)^2 + 2 \right]$$

2.

$$\begin{aligned}
 \frac{dw}{dx} &= \frac{dw}{dy} \cdot \frac{dy}{dx} \\
 &= 2ay(2bx + c) \\
 &= 2a(bx^2 + cx)(2bx + c)
 \end{aligned}$$

3. (a) Let $z = 3x^2 - 13$, so $y = z^3$:

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = 3z^2(6x) = 18x(3x^2 - 13)^2$$

(b) Let $z = 7x^3 - 5$, so $y = z^9$:

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = 9z^8 (21x^2) = 189x^2 (7x^3 - 5)^8$$

(c) Let $z = ax + b$, so $y = z^5$:

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = 5z^4(a) = 5a(ax + b)^4$$

4. Chain rule, with $z = 16x + 3$ and $y = z^{-2}$:

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = -2z^{-3}(16) = \frac{-32}{(16x + 3)^3}$$

Quotient rule, where the derivative of the denominator is $2(16x+3) \cdot 16 = 32(16x+3)$:

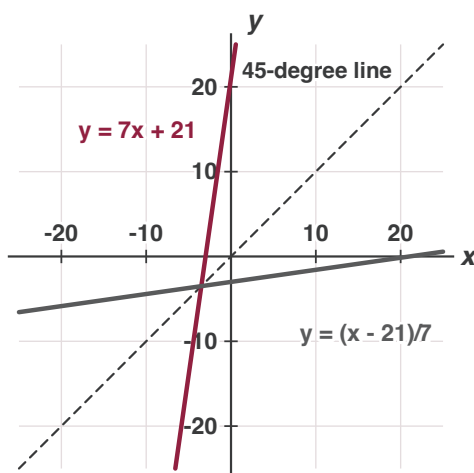
$$\frac{dy}{dx} = \frac{0 \cdot (16x + 3)^2 - 1 \cdot 32(16x + 3)}{(16x + 3)^4} = \frac{-32}{(16x + 3)^3}$$

The answers are identical.

5. The inverse function is $x = \frac{y-21}{7}$.

$$\frac{dy}{dx} = 7 \quad \frac{dx}{dy} = \frac{1}{7} = \frac{1}{dy/dx}$$

so the inverse-function rule holds. To graph the inverse with its own independent variable on the horizontal axis, write it as $y = (x - 21)/7$. The two lines are mirror images of each other across the 45-degree line $y = x$:



6. (a)

$$\frac{dy}{dx} = -6x^5 < 0 \text{ for } x > 0$$

So the function is strictly decreasing, hence strictly monotonic, and

$$\frac{dx}{dy} = \frac{1}{dy/dx} = \frac{-1}{6x^5}$$

(b)

$$\frac{dy}{dx} = 20x^4 + 3x^2 + 3 > 0 \text{ for all } x$$

So the function is strictly increasing, hence strictly monotonic, and

$$\frac{dx}{dy} = \frac{1}{20x^4 + 3x^2 + 3}$$

Exercise 10.5

1. (a) $\frac{dy}{dt} = 2e^{2t+4}$

(b) $\frac{dy}{dt} = -9e^{1-9t}$

(c) $\frac{dy}{dt} = 2te^{t^2+1}$

(d) $\frac{dy}{dt} = -10te^{2-t^2}$

$$(e) \frac{dy}{dx} = (2ax + b)e^{ax^2+bx+c}$$

$$(f) \frac{dy}{dx} = x \frac{d}{dx} e^x + e^x \frac{d}{dx} x = xe^x + e^x = (x+1)e^x$$

$$(g) \frac{dy}{dx} = x^2 (2e^{2x}) + 2xe^{2x} = 2x(x+1)e^{2x}$$

$$(h) \frac{dy}{dx} = a (xbe^{bx+c} + e^{bx+c}) = a(bx+1)e^{bx+c}$$

$$3. (a) \frac{dy}{dt} = \frac{35t^4}{7t^5} = \frac{5}{t}$$

$$(b) \frac{dy}{dt} = \frac{10at^9}{at^{10}} = \frac{10}{t}$$

$$(c) \frac{dy}{dt} = \frac{1}{t+19}$$

$$(d) \frac{dy}{dt} = 5 \cdot \frac{2(t+1)}{(t+1)^2} = \frac{10}{t+1}$$

$$(e) \frac{dy}{dx} = \frac{1}{x} - \frac{1}{1+x} = \frac{1}{x(1+x)}$$

$$(f) \frac{dy}{dx} = \frac{d}{dx} [\ln x + 8 \ln(1-x)] = \frac{1}{x} - \frac{8}{1-x} = \frac{1-9x}{x(1-x)}$$

$$(g) \frac{dy}{dx} = \frac{d}{dx} [\ln 2x - \ln(1+x)] = \frac{2}{2x} - \frac{1}{1+x} = \frac{1}{x(1+x)}$$

$$(h) \frac{dy}{dx} = 5x^4 \cdot \frac{2x}{x^2} + 20x^3 \ln x^2 = 10x^3 (1 + 2 \ln x^2) = 10x^3 (1 + 4 \ln x)$$

$$7. (a) \text{ Since } \ln y = \ln 3x - \ln(x+2) - \ln(x+4),$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x} - \frac{1}{x+2} - \frac{1}{x+4} = \frac{8-x^2}{x(x+2)(x+4)}$$

Hence,

$$\frac{dy}{dx} = \frac{8-x^2}{x(x+2)(x+4)} \cdot \frac{3x}{(x+2)(x+4)} = \frac{3(8-x^2)}{(x+2)^2(x+4)^2}$$

$$(b) \text{ Since } \ln y = \ln(x^2+3) + x^2 + 1,$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{2x}{x^2+3} + 2x = \frac{2x(x^2+4)}{x^2+3}$$

Hence,

$$\frac{dy}{dx} = \frac{2x(x^2+4)}{x^2+3} \cdot (x^2+3)e^{x^2+1} = 2x(x^2+4)e^{x^2+1}$$